

# Yoel Kim

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## Research Interests

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- Software Engineering → software verification; specification mining; embedded software.
- Formal Methods → abstraction; automata learning; model checking.
- Programming Languages → program analysis; program synthesis.

## Education

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- Ph.D. in Computer Science and Engineering, Kyungpook National University* *Mar 2023 – Feb 2027  
(Expected)*
- Advisor: Yunja Choi
  - Thesis: Data-Driven Abstractions for Embedded Software
- M.S. in Computer Science and Engineering, Kyungpook National University* *Mar 2021 – Feb 2023*
- Advisor: Yunja Choi
  - Thesis: An Automated Stub Generation Approach Using Program Synthesis for Software Verification
- B.S. in Computer Science and Engineering, Kyungpook National University* *Mar 2017 – Feb 2021*
- Track: Global Software Convergence
  - GPA: 3.95/4.3, Rank: 1/13

## Ongoing Work

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1. AUTOCL: An End-to-End Tool for Active Automata Learning from C Programs  
Yoel Kim, Yong-Hwan Jeong, and Yunja Choi  
(In preparation) **TACAS'27: International Conference on Tools and Algorithms for the Construction and Analysis of Systems** (Regular Tool Papers Track)
2. Learning Functional Specifications from Stateful Control Programs through Active Model Learning  
Yunja Choi and Yoel Kim  
(Submitted) *IEEE Access*, 2026

## International Conferences

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1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper  
Yoel Kim and Yunja Choi  
**FSE'26: ACM International Conference on the Foundations of Software Engineering**
2. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software  
Yoel Kim and Yunja Choi  
**FSE'24: ACM International Conference on the Foundations of Software Engineering**

## Domestic Conferences

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1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation  
Seungbin Choi, Yoel Kim, and Yunja Choi  
KCSE'26: *Korea Conference on Software Engineering* \* *Short Paper Award*
2. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation  
Yoel Kim and Yunja Choi  
KCSE'23: *Korea Conference on Software Engineering* \* *Best Short Paper Award*
3. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis  
Yoel Kim and Yunja Choi  
KSC'21: *Korea Software Congress*

4. A Case Study on the Performance of Program Synthesis in Embedded Software Domain  
Yoel Kim and Yunja Choi  
*KCSE'21: Korea Conference on Software Engineering*

## Domestic Journals

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1. An Analysis of the Feasibility and Limitations of LLM-based State Machine Generation for Reactive Systems  
Seungbin Choi, Yoel Kim, and Yunja Choi  
*KIPS Transactions on Software and Data Engineering*, 2026

## Selected Talks

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1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper. Research Paper Presentation. FSE'26. *Concordia University, Montreal, Canada. Jul 7, 2026.*
2. AUTOCL: Active Learning of Symbolic Automata via Dynamic Symbolic Mapper. Milestone Talk. 10th STAAR Workshop. *Ulsan, Korea. Feb 3, 2026.*
3. C2FSM: A Tool for Active Learning of Extended Finite State Machines from C Programs. Milestone Talk. 9th STAAR Workshop. *Korea University, Seoul, Korea. Jul 29, 2025.*
4. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCC'25. *Jeju, Korea. Jul 4, 2025.*
5. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCSE'25. *Pyeongchang, Korea. Jan 22, 2025.*
6. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Research Paper Presentation. FSE'24. *Porto de Galinhas, Ipojuca, Pernambuco, Brazil. Jul 17, 2024.*

## Experience

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- Teaching Assistant**, Kyungpook National University *Mar 2021 – Jun 2024*
- ITEC0414: Software Testing Theory (Spring 2022, 2023, 2024).
  - COMP0224: Software Design (Fall 2021, 2022).
  - COMP0216: Data Structure Applications (Spring 2021).
- Undergraduate Tutor**, Kyungpook National University *Mar 2018 – Dec 2020*
- COMP0322: Database Management Systems (Fall 2020).
  - CLTR266: Software Computational Thinking (Spring 2018, 2019).
  - CLTR268: Python Programming (Winter 2018).
  - COMP0216: Data Structure Applications (Fall 2018).
- Undergraduate Research Intern**, Software Safety Engineering Lab *Sep 2020 – Feb 2021;  
Jul 2019 – Dec 2019*
- Supervisor: Yunja Choi
  - Developed a Java-based GUI tool to support software testing activities for Hyundai Motor.
- Intern**, JS System, Daegu *Dec 2019 – Feb 2020*
- Assisted with testing an FPGA-based embedded board.
- Undergraduate Research Intern**, Intelligent Network Lab *Oct 2018 – Dec 2018*
- Supervisor: Dongkyun Kim